

Orthogonal Translation Computed Laminography Reconstruction Based on Self-Prior Information and Adaptive Weighted Total Variation

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Abstract—Orthogonal translation computed laminography (OTCL) provides an effective non-destructive testing technique for plate-like objects. Nevertheless, the reconstructed OTCL images always suffer from aliasing artifacts due to the incompleteness of projection data, which have a negative impact on defect judgment, dimension measurement and failure analysis. To investigate the causes of aliasing artifacts, the spatial distribution characteristics of OTCL within the three-dimensional frequency domain were analyzed. Subsequently, a novel reconstruction OTCL algorithm was proposed, termed self-prior information guidance and adaptive weight total variation constraint (SPIG-AwTV). The SPIG-AwTV is designed to mitigate aliasing artifacts with a self-prior information guidance (SPIG) regularization term and an adaptive weighted total variation (AwTV) regularization term. The SPIG regularization term is derived from filtered backprojection reconstruction result through contour extraction and mask operation. The AwTV regularization term is tailored based on the gradient features of OTCL images. Comparing with the latest methods, experimental results demonstrate that the SPIG-AwTV has the greater capacity to suppress aliasing artifacts, preserve edges, and thereby obtain higher quality OTCL images.

Index Terms—Computed laminography; Self-prior information; Total variation; Image reconstruction.

I. INTRODUCTION

Computed tomography (CT) employs the penetrability of X-rays and a scanning scheme involving relative rotation between the object and the source-detector to attain imaging [1], [2]. Nevertheless, when imaging plate-like objects, CT will face the challenges of being difficult to penetrate in width

direction and mechanical structure collisions [3], [4]. On the basis of CT scan, a new imaging method is formed by simultaneously tilting the workpiece at a certain angle with the rotation centerline: rotational computed laminography (RCL) [5], [6], [7]. RCL overcomes the issue of penetration and mechanical collisions, and achieves a uniform change in penetration thickness [8]. However, the limited distribution angle of the source necessitates the tilting of the source during scanning, making it impossible for RCL to achieve large geometric magnification.

Orthogonal translation computed laminography (OTCL) [9] is an easy to implement method for computed laminography (CL), which overcomes the shortcomings of RCL. In OTCL scanning, the source can be positioned as close as possible to the plate-like object, enabling high magnification and high-resolution imaging. However, OTCL also only collects the projection data from the thickness direction, resulting in incomplete projection data. Consequently, OTCL image reconstruction belongs to an underdetermined inverse problem, the reconstructed images will exhibit severe aliasing artifacts and a substantial decrease in contrast [10], [11].

Suppressing aliasing artifacts and restoring image details has become one of the focuses in computed laminography (CL) image reconstruction [12]. Although deep learning methods have been widely studied in solving inverse problem [13], the special imaging method of CL makes it difficult to meet the accurate reconstruction data conditions, resulting in difficulties in obtaining label. Therefore, current research mainly focuses on CL iterative reconstruction algorithms. These methods involve the integration of prior knowledge into iterative algorithms, which effectively mitigate the issue of ill-posed reconstructions resulting from incomplete projection data [14], [15]. Prior information employed to mitigate aliasing artifacts can be categorized into two main types: the image gradient sparsity prior and the structural prior [16].

Based on the image gradient sparsity prior, the total variation (TV) regularization term can be constructed, which effectively reduces aliasing artifacts by minimizing the gradient of the image. However, traditional TV regularization term may result in overly smooth edges [17], [18]. To mitigate this problem, Liu Yuhang et al. [19] proposed a truncated adaptive weight TV (TAwTV) regularization term, and employed a simultaneous algebraic reconstruction technique-truncated adaptive weight TV (SART-TAwTV) framework to

¹Manuscript received xx 2024; revised xx 2024; accepted xx 2024. Date of publication xx 2024; date of current version xx 2024. This work was supported by the National Key Research and Development Program of China (No. 2022YFF0706400); The Associate Editor coordinating the review process was xxx. (Corresponding authors: Yufang Cai; Liming Duan.) Chuangdong Tan, Chao Long, Yarui Xi, Xinxin Lin, Zhiting Chen, Fenglin Liu, Yufang Cai, and Liming Duan are with the Key Laboratory of Optoelectronic Technology and Systems, Ministry of Education, Chongqing University, Chongqing 400044. (e-mail: caiyf@cqu.edu.cn, duanliming@cqu.edu.cn).

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solve the TAwTV. The TAwTV penalizes only gradients below a specified threshold, thereby alleviating edge transition smoothing. Additional regularizations include the anisotropic alternating TV (AARTV) [20] and adaptive weighted dynamic adjusted relative TV (AwDaRTV) regularization [21]. Despite achieving satisfactory image quality, the TV regularization-based reconstruction algorithms still face some challenges, such as the complexity in adjusting algorithm parameters.

Structural priors typically encompass computer-aided design (CAD) file, or grayscale information [22], [23]. Christian Schorr et al. [24] employed a CAD as prior information and developed a semi-automatic prior image registration algorithm, thereby mitigating artifacts and enhancing contrast. **The above method increases the protection of structural features, but heavily relies on prior information and involves image registration issues.** The discrete algebraic reconstruction technique (DART) leverages the grayscale of the object as prior information, which enhances the quality of images. Nonetheless, this method is limited to detecting objects composed of fewer material types [25].

At present, there are also methods that combine the above two types of prior information and propose the concept of hybrid prior. Hybrid prior combine different types of prior knowledge to better capture the complex characteristics of images, thereby improving the quality of reconstructed images. Such as, Xi Yarui et al. [16] proposed a prior information guided directional total p-variation algorithm, which improved the image quality of OTCL local imaging. Gui Zhiguo et al. [26] proposed a prior-image based reconstruction algorithm (PIBR) for plate-like object image reconstruction. PIBR acquires prior information by two full scans of standard workpieces at different tilt angles, the complexity of obtaining prior information is notable. The drawback is that the above method also requires precise registration.

To mitigate the limitations of current algorithms that struggle to suppress aliasing artifacts in OTCL images, we analyzed the spatial distribution characteristics of OTCL within the three-dimensional (3D) frequency domain. It can be found that OTCL images possess a wealth of high-frequency information, while low-frequency data are missing. This abundance of high-frequency information contributes to rich contour detail in OTCL images. Capitalizing on this observation, we proposed a self-prior information guidance and adaptive weight total variation (AwTV) constraint algorithm (termed SPIG-AwTV) to suppress aliasing artifacts and enhance the reconstructed image quality. The primary contributions of this work lie in the following aspects:

1) The 3D frequency domain characteristics of OTCL under parallel beam conditions was studied, and proved that the aliasing artifacts and low contrast in OTCL images are due to the lack of pyramid shaped data that extends up and down from the center slice as the vertex position in the 3D frequency domain of OTCL. This finding provides theoretical support for the design of OTCL image reconstruction algorithms.

2) A self-prior information guidance (SPIG) regularization term is derived from CL filtered backprojection

(CLFBP) reconstruction result through contour extraction and mask operation. The SPIG regularization term adds image constraints to the X-Y plane, and suppresses aliasing artifacts in the reconstructed image.

3) An AwTV regularization term was developed, which adaptively adjusts the degree of constraint on different region based on gradient. This adaptation reduces the constraints on regions with higher gradient values, while enhancing the constraints in the region with lower gradient values. **Accordingly, the edges will be better preserved and the reconstructed images will be more accurate.**

4) A SPIG-AwTV reconstruction algorithm for OTCL was proposed by combining the SPIG regularization term and the AwTV regularization term. This comprehensive approach to reconstruction constrains the OTCL results from various perspectives, thereby significantly improving the quality of the reconstructed images.

In summary, this study proposed an innovative solution for mitigating aliasing artifacts in OTCL images and achieving superior image quality, which is more conducive to the subsequent implementation of defect recognition [27], structural analysis [28], reverse design, etc. **In contrast to extant reconstruction methods, SPIG-AwTV obviates the need for prior information acquisition and the subsequent registration of prior information and reconstructed images.**

The structure of this article is as follows. Section II introduces the OTCL scan and associated reconstruction algorithms. Section III describes the proposed reconstruction method. The numerical experiments and physical practical experiments are presented in section IV. In section V, we give a brief discussion and conclusion of our research work.

II. RELATED WORKS

A. OTCL imaging model

The scanning geometry of OTCL is illustrated in Fig. 1. During the OTCL imaging, the source emits X-rays and the detector receives the attenuated X-rays transmitted through the plate-like object. The plate-like object is fixed, the X-ray source and detector move relative to each other along the X-axis to complete the scanning in the x-direction. Subsequently, the object undergoes a 90° rotation around the Z-axis, after which the source and detector execute relative translation movements to complete the scanning in the y-direction [9].

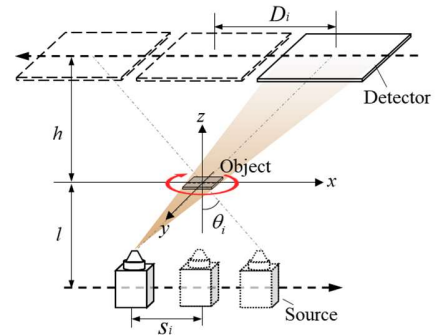


Fig. 1. Schematic diagram of OTCL scanning.

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A 3D geometric coordinate system is established in Fig.1, with the origin located at the center of the plate-like object. Based on the vertical distance l from the source to the object, and the vertical distance h from the detector to the object, the motion coordinates of the source $(S_i, 0, -l)$ and the detector $(D_i, 0, h)$ can be calculated. Among them, S_i and D_i satisfy the geometric relationships of (1) - (2) respectively:

$$S_i = l \times \tan(\theta_i) \quad (1)$$

$$D_i = h \times \tan(-\theta_i) \quad (2)$$

where, θ_i represents the angle between the direction of the central ray and the Z-axis. Generally, $\theta_i = [-\pi/4, \pi/4]$.

B. Analytical algorithm for OTCL

OTCL image reconstruction algorithms are categorized into analytical algorithms and iterative algorithms. The filtered backprojection (FBP) algorithm, serves as a prominent example of an analytical reconstruction algorithm. The relationship between the reconstructed image and the corresponding projection data recorded by the detector can be mathematically represented as follows:

$$Af = p \quad (3)$$

where, A represents the projection matrix or system matrix. Define $T_{\theta,z}(x, y)$ as the reconstructed image under a projection angle θ_i and a layered position z , having:

$$T_{\theta,z}(x, y) = \mathbf{BP}_{\theta,z}(p_{\bar{\theta}}(\vec{s})) \quad (4)$$

$p_{\bar{\theta}}(\vec{s})$ is the filtered projection under angle θ_i , $\mathbf{BP}(\bullet)$ represents the backprojection operator, and the process of reconstructing $\hat{f}$ by backprojection can be defined as the weighted accumulation of all backprojection perspective maps:

$$\hat{f} = \sum_{\forall \theta_i \in \Gamma} w_{u,v}(x, y, z, \theta_i) \mathbf{BP}_{\theta,z}(p_{\bar{\theta}}(\vec{s})) \quad (5)$$

where, $w_{u,v}(x, y, z, \theta_i)$ represents the backprojection weighting coefficient of the projection $p_{\bar{\theta}}(\vec{s})$, which is generally obtained through linear interpolation.

The analytical algorithms of CL can be delineated into three steps: weighting, filtering, and backprojection. Based on this idea, Zhang Yanan et al. [29] proposed an FBP reconstruction algorithm for RCL. Cai Yufang et al. [30] proposed FBP algorithms for OTCL.

C. Iterative algorithms for OTCL

The iterative reconstruction of OTCL can be formulated as solving Eq. (3). Mathematically, the solution of Eq. (3) is to recover the unknown image f from the known system matrix A and projection data p , which can be solved through the least squares problem. This involves solving the optimization problem as described in Eq. (6):

$$\arg \min_f \|Af - p\|_2^2 \quad (6)$$

OTCL image reconstruction is a type of inverse problem

involving incomplete projection data. Consequently, prior information is typically employed to augment constraints during the reconstruction process. TV regularization is a prevalent technique in CL image reconstruction [31], as it leverages the sparse gradient information of the CL images. The objective function for the associated reconstruction algorithm is defined as follows:

$$\arg \min_f \|Af - p\|_2^2 + \lambda \|f\|_{TV} \quad (7)$$

where, λ is the balance parameter; $\|f\|_{TV}$ is TV regularization term can be solved by gradient descent method [32] or split bregman (SB) method [33].

III. THEORY AND METHOD

A. OTCL frequency domain analysis

To analyze the causes of aliasing artifacts in OTCL image, we analyzed the 3D frequency domain distribution of OTCL. Define the 3D objective function as $f(\vec{x})$ and $\vec{x} = [x, y, z]^T$. As shown in Fig. 2(a), the projection of the 3D function $f(\vec{x})$ along angle θ_i direction is:

$$p_{\bar{\theta}}(\vec{s}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_3(\vec{x}) \delta(N^T \vec{x} - \vec{s}) d^3 \vec{x} \quad (8)$$

where $\vec{s} = [s, t]^T$, δ is the Dirac function, N represents the projection normal vector matrix:

$$N = \begin{bmatrix} \sin \varphi_i & \cos \varphi_i \cos \theta_i \\ -\cos \varphi_i & \sin \varphi_i \cos \theta_i \\ 0 & -\sin \varphi_i \end{bmatrix} \quad (9)$$

where, φ_i represents the unit space vector along the projection direction, and θ_i represents the angle between the X-ray direction and the Z-axis. For OTCL scanning $\theta_i = [-\pi/4, \pi/4]$, $\varphi_i = \{0, \pi/2, \pi, 3\pi/2\}$. After two dimensional Fourier transform for $p_{\bar{\theta}}(\vec{s})$, we can obtain:

$$\mathcal{F} p_{\bar{\theta}}(\vec{\omega}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p_{\bar{\theta}}(\vec{s}) \exp(-j2\pi \vec{\omega}^T \vec{s}) d^2 \vec{s} \quad (10)$$

the corresponding plane of $\mathcal{F} p_{\bar{\theta}}(\vec{\omega})$ is represented as $\mathcal{F}(ku, kv)$, and the angle between the plane and ω_z is $\pi/2 - \theta_i$. The OTCL imaging θ_i constantly changes, the angle $\pi/2 - \theta_i$ between $\mathcal{F}(ku, kv)$ and ω_z also changes. The final 3D frequency domain distribution is shown in Fig. 2 (b).

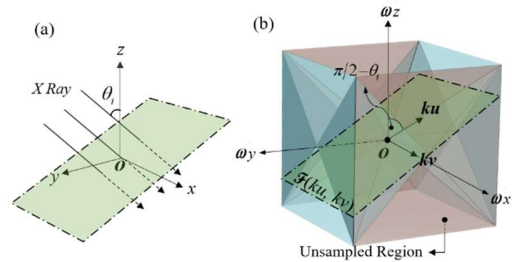


Fig. 2. OTCL frequency domain distribution, (a) projection diagram, (b) distribution of frequency domain of OTCL.

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Obviously, OTCL inherently faces challenges with missing data in the Fourier domain, making precise reconstruction of OTCL images theoretically unattainable. This issue arises from a pyramid-shaped void that extends vertically from the central slice within the 3D Fourier space, characteristic of the laminography geometry. In contrast to alternative techniques, OTCL effectively integrates structural details along both the X and Y directions, yielding a more extensive collection of image information in the frequency domain.

Analyzing the frequency domain information of OTCL along the w_x - w_y plane, it reveals a deficiency in low-frequency components within the frequency domain space. Low frequency elements correspond to the grayscale content, whereas high frequency components capture intricate details such as contours. Consequently, the OTCL image reconstructed by analytical algorithms may lead to contrast degradation, the emergence of aliasing artifacts and other related issues. Nonetheless, owing to the abundance of high-frequency data in the 3D frequency domain space of OTCL, it is feasible to essentially restore contour information. Inspired by this characteristic, we proposed a method to obtain contour information of OTCL images from CLFBP reconstructed results, and further forms a self-prior regularization term.

B. Proposed SPIG-AwTV

To mitigate aliasing artifacts in OTCL images, we proposed the SPIG-AwTV algorithm. SPIG-AwTV includes a **SPIG regularization term** and an AwTV regularization term.

Definition: the optimization model of SPIG-AwTV algorithm can be formulated as:

$$\arg \min_f \left\{ \frac{1}{2} \|Af - p\|_2^2 + \lambda_1 \|f - (M_{fill} f_{edge})\|_2^2 + \|f\|_{AwTV} \right\} \quad (11)$$

Among them, the first term signifies the data fidelity aspect, the second term denotes the SPIG regularization term, and the third term corresponds to the AwTV regularization term. λ_1 is a balance parameter of SPIG, which influences the severity of the prior information penalty. The weight of AwTV regularization term is set to 1. Due to its coupling with the hyperparameters of AwTV, weight adjustment of AwTV can be achieved through its hyperparameters.

The SPIG regularization term utilizes contour information to increase the protection of structural features. The AwTV regularization term adaptively enhances the constraints on structures exhibiting smaller gradient values and reduces the constraints on structures with larger gradient values based on the gradient differences of OTCL images. Under the constraints of the two regularization terms, it is possible to protect image edges and mitigate excessive smoothing during the OTCL image reconstruction.

As depicted in Fig. 3, the reconstruction process of SPIG-AwTV can be divided into three stages: projection acquisition, self-prior extraction and OTCL image reconstruction.

(a) Projection acquisition: Firstly, the OTCL projection data is carried out along the x -direction, with the projection data represented as p_x . Then, the data collection is carried out along the y -direction, with the projection data represented as p_y .

(b) Self-prior information extraction: the CLFBP algorithm is used to reconstruct 3D images of f_1 and f_2 from the projection data p_x and p_y , respectively. The two-dimensional images in the X-Y plane are represented as f_1^{xy} and f_2^{xy} . It can be observed that the structure of f_1^{xy} in the x -direction is clearly visible, while the structure of f_2^{xy} in the y -direction is clearly visible. This phenomenon is caused by the special scanning method of OTCL. Therefore, the edges of the X-Y plane can be calculated by Eq. (12):

$$f_{edge} = \sqrt{\left(\partial(f_1^{xy})/\partial x\right)^2 + \left(\partial(f_2^{xy})/\partial y\right)^2} \quad (12)$$

The edge filling operator $M_{fill}(\bullet)$ shown in Eq. (13) is used to fill the edges, thus forming $M_{fill} f_{edge}$. It should be noted that $M_{fill} f_{edge}$ is a binary image.

$$M_{fill}(\bullet) = \begin{cases} 1, & \in \text{in edge} \\ 0, & \in \text{out edge} \end{cases} \quad (13)$$

(c) OTCL image reconstruction: OTCL image reconstruction is an underdetermined inverse problem. To improve the quality of reconstructed images, this paper designs two regularization terms and introduces them into the reconstruction process. Firstly, mask the reconstruction result f of the last iteration with $M_{fill} f_{edge}$ and obtain the self-prior image $(M_{fill} f_{edge})f$, calculate the difference between the reconstructed image f and $(M_{fill} f_{edge})f$, and form the self-prior regularization term.

C. Solving algorithm

SPIG-AwTV incorporates both L_1 and L_2 norm, which cannot be directly differentiated to obtain the optimal value. For the convenience of solving, we refer to references [26] and [34] to decompose the SPIG-AwTV into two subproblems: image reconstruction guided by self-prior information and AwTV constrained image optimization.

Subproblem-1: Image reconstruction guided by self-prior information

$$\tilde{f} = \arg \min_f \left\{ \frac{1}{2} \|Af - p\|_2^2 + \lambda_1 \|f - (M_{fill} f_{edge})f\|_2^2 \right\} \quad (14)$$

Using the simultaneous algebraic reconstruction techniques (SART) to solve the subproblem of self-prior information guided reconstruction, the iterative expression is as follows:

$$\tilde{f} = (A^T A + \lambda_1 I)^{-1} (A^T p + \lambda_1 M_{fill} f_{edge}) \quad (15)$$

$$\begin{aligned} \tilde{f}_j^{l+1} &= \tilde{f}_j^l + \lambda (1 / \sum_{i=1}^N a_{ij}) \\ &\times \left[\sum_{i=1}^N a_{ij} (p_i - \sum_{i=1}^N a_{ij} \tilde{f}_j^l) \frac{1}{\sum_{i=1}^N a_{ij}} + \lambda_1 ((M_{fill} f_{edge}) \tilde{f}_j^l - \tilde{f}_j^l) \right] \end{aligned} \quad (16)$$

where, λ is the relaxation factor ($0 < \lambda < 2$), I represents the identity matrix, $\tilde{f}_j^l$ is the pixel value of the j th pixel in the l th iteration, p_i represents the projection value of the i th ray, a_{ij} represents the contribution of the j th pixel to the i th ray.

Subproblem-2: AwTV constrained image optimization

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$$\hat{f} = \arg \min_f \left\{ \frac{\mu}{2} \|f - \tilde{f}\|_2^2 + \|f\|_{AwTV} \right\} \quad (17)$$

where, μ is a weight parameter, AwTV is defined as:

$$\|f\|_{AwTV} = \sum_{x,y,z} \sqrt{\omega_{x,y,z}^x (\nabla_x f)^2 + \omega_{x,y,z}^y (\nabla_y f)^2 + \omega_{x,y,z}^z (\nabla_z f)^2} \quad (18)$$

where $\nabla_x f, \nabla_y f, \nabla_z f$ represent the gradients of the image along the x, y and z directions. **The adaptive weights $\omega_{x,y,z}^x, \omega_{x,y,z}^y, \omega_{x,y,z}^z$ are introduced to the regularization term in different directions. In this way, a stronger weight is given for the smaller gradient of voxel intensity and a weaker one for the larger gradient of voxel intensity.**

The adaptive weight function can be denoted as:

$$\omega_{x,y,z}^x = e^{-\frac{-(\nabla_x f)^2}{\sigma^2}}, \omega_{x,y,z}^y = e^{-\frac{-(\nabla_y f)^2}{\sigma^2}}, \omega_{x,y,z}^z = e^{-\frac{-(\nabla_z f)^2}{\sigma^2}} \quad (19)$$

σ is a weight control parameter, **which is related to gradient value. Choosing a suitable δ for AwTV is an essential task.** If σ is too small, the overall weight will be too small, and the suppression effect of artifacts will be poor. If σ is too large, the weight value will approach to 1, making the AwTV similar to the traditional TV.

Eq. (17) contains L_1 and L_2 norms, and cannot be directly differentiated. The SB framework is used to solve the AwTV minimization problem. Firstly, introduce three auxiliary variables d_x, d_y, d_z to transform Eq. (17) into Eq. (20):

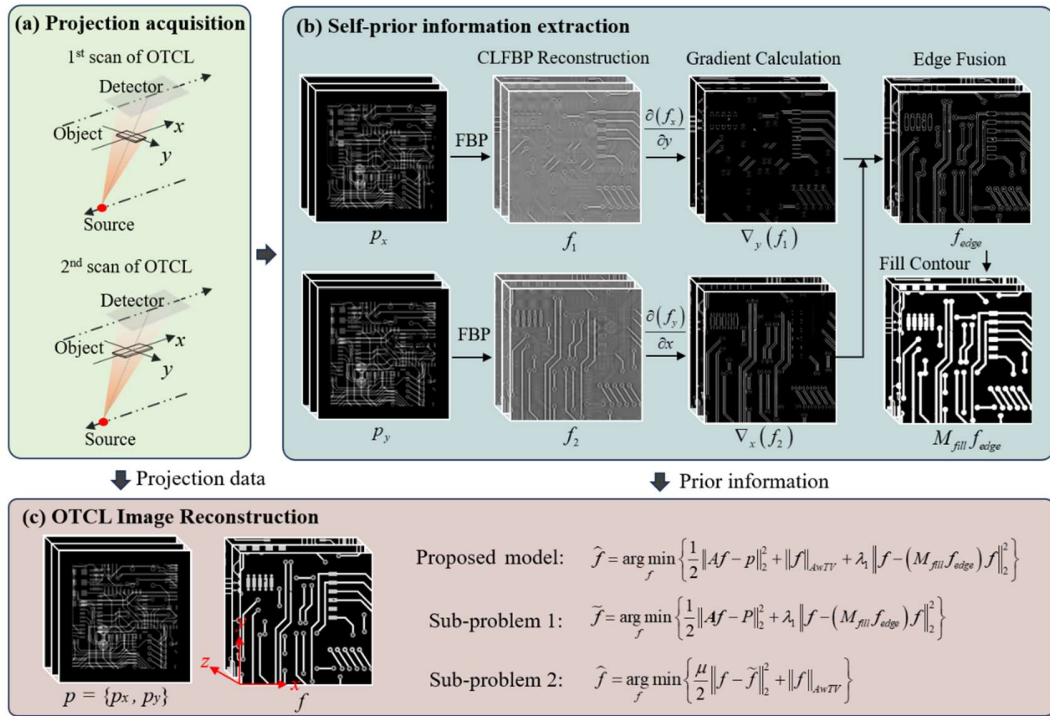


Fig. 3. Overall flow for OTCL reconstruction by the proposed SPIG-AwTV algorithm, which includes three stages: projection acquisition, self-prior information extraction and OTCL image reconstruction.

$$\arg \min_f \left\{ \omega_{x,y,z}^x \|d_x\|_1 + \omega_{x,y,z}^y \|d_y\|_1 + \omega_{x,y,z}^z \|d_z\|_1 + \frac{\mu}{2} \|f - \tilde{f}\|_2^2 \right\}, \quad s.t. \quad d_x = \nabla_x f, d_y = \nabla_y f, d_z = \nabla_z f \quad (20)$$

transform Eq. (20) into an unconstrained problem as Eq. (21):

$$\arg \min_f \left\{ \omega_{x,y,z}^x \|d_x\|_1 + \omega_{x,y,z}^y \|d_y\|_1 + \omega_{x,y,z}^z \|d_z\|_1 + \frac{\mu}{2} \|f - \tilde{f}\|_2^2 + \frac{\alpha}{2} \|d_x - \nabla_x f\|_2^2 + \frac{\alpha}{2} \|d_y - \nabla_y f\|_2^2 + \frac{\beta}{2} \|d_z - \nabla_z f\|_2^2 \right\} \quad (21)$$

α, β are balanced variables. Eq. (22) and Eq. (23) can be obtained through the SB frame:

$$\left\{ \hat{f}^{k+1}, d_x^{k+1}, d_y^{k+1}, d_z^{k+1} \right\} = \min \left\{ \begin{aligned} & \frac{\mu}{2} \|f - \tilde{f}\|_2^2 + \omega_{x,y,z}^x \|d_x\|_1 + \omega_{x,y,z}^y \|d_y\|_1 + \omega_{x,y,z}^z \|d_z\|_1 \\ & + \frac{\alpha}{2} \|d_x - \nabla_x f - b_x^k\|_2^2 + \frac{\alpha}{2} \|d_y - \nabla_y f - b_y^k\|_2^2 + \frac{\beta}{2} \|d_z - \nabla_z f - b_z^k\|_2^2 \end{aligned} \right\} \quad (22)$$

$$b_x^{k+1} = b_x^k + \nabla_x f^{k+1} - d_x^{k+1}, \quad b_y^{k+1} = b_y^k + \nabla_y f^{k+1} - d_y^{k+1}, \quad b_z^{k+1} = b_z^k + \nabla_z f^{k+1} - d_z^{k+1} \quad (23)$$

Split the L_1 and L_2 norm components of Eq. (22) to obtain Eq. (24) and Eq. (25), and use the alternating direction method of multipliers to iteratively update $\hat{f}, dx, dy, dz$.

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$$\hat{f}^{k+1} = \min \left\{ \begin{aligned} &\frac{\mu}{2} \|f - \tilde{f}\|_2^2 + \frac{\alpha}{2} \|d_x - \nabla_x f - b_x^k\|_2^2 \\ &+ \frac{\alpha}{2} \|d_y - \nabla_y f - b_y^k\|_2^2 + \frac{\beta}{2} \|d_z - \nabla_z f - b_z^k\|_2^2 \end{aligned} \right\} \quad (24)$$

$$\begin{aligned} d_x^{k+1} &= \min \left\{ \omega_{x,y,z}^x \|d_x\|_1 + \frac{\alpha}{2} \|d_x - \nabla_x f^{k+1} - b_x^k\|_2^2 \right\} \\ d_y^{k+1} &= \min \left\{ \omega_{x,y,z}^y \|d_y\|_1 + \frac{\alpha}{2} \|d_y - \nabla_y f^{k+1} - b_y^k\|_2^2 \right\} \\ d_z^{k+1} &= \min \left\{ \omega_{x,y,z}^z \|d_z\|_1 + \frac{\beta}{2} \|d_z - \nabla_z f^{k+1} - b_z^k\|_2^2 \right\} \end{aligned} \quad (25)$$

Eq. (24) is differentiable, so the Gauss-Seidel method [35] is used to solve $\hat{f}$. The optimal value of $\hat{f}$ can be solved by taking the derivative of Eq. (24) and setting the result to zero, and then using twice Fourier transform:

$$\hat{f} = \mathcal{F}^{-1} \left(\frac{\mu \mathcal{F}(\tilde{f}) - \mathcal{F}(\text{div}(d^k - b^k))}{\mu \mathcal{F}(I) - \mathcal{F}(\Delta)} \right) \quad (26)$$

where, $\mathcal{F}(\cdot)$ represents Fourier transform operator, $\mathcal{F}^{-1}(\cdot)$ stands for inverse Fourier transform, $\Delta(\cdot)$ is Laplace operator, $\text{div}(\cdot)$ presents the divergence operator, $I = A^T A$, $d^k = (\alpha d_x^k, \alpha d_y^k, \beta d_z^k)^T$, $b^k = (\alpha b_x^k, \alpha b_y^k, \beta b_z^k)^T$. Eq. (25) is solved by the contraction formula [4], and the result is shown in Eq. (27):

$$\begin{aligned} d_x^{k+1} &= \text{shrink}(\nabla_x f^{k+1} + b_x^k, \frac{\omega_{x,y,z}^x}{\alpha}) \\ d_y^{k+1} &= \text{shrink}(\nabla_y f^{k+1} + b_y^k, \frac{\omega_{x,y,z}^y}{\alpha}) \\ d_z^{k+1} &= \text{shrink}(\nabla_z f^{k+1} + b_z^k, \frac{\omega_{x,y,z}^z}{\beta}) \end{aligned} \quad (27)$$

where, $\text{shrink}(\chi, \gamma) = \chi / |\chi| \times \max(|\chi| - \gamma, 0)$.

The pseudo code of the proposed SPIG-AwTV algorithm is listed in Algorithm 1.

Algorithm 1: Pseudo code of SPIG-AwTV algorithm

1. **Initial:** $\lambda, \lambda_1, dx, dy, dz, a, \beta, \mu, \sigma$
2. **For** $t = 1 : M_{iter}$
3. **For** $l = 1 : R_{iter}$ (Step 1: SPIG Loop)
4. Update image $\tilde{f}^l$ by Eq. (16)
5. **End**
6. **For** $k = 1 : S_{iter}$ (Step 2: AwTV Loop)
7. Update $\hat{f}^k$ by Eq. (26)
8. Update d_x^k, d_y^k, d_z^k by Eq. (27)
9. Update b_x^k, b_y^k, b_z^k by Eq. (23)
10. **End**
11. **End**
12. **Output:** $\hat{f}$

Among them, M_{iter} represents the total number of iterations of the algorithm, R_{iter} represents the number of SPIG iterations that introduce priors, and S_{iter} represents the number of

iterations of AwTV. The initial values of dx, dy and dz are all set to 0. According to the pseudocode of our Algorithm 1, the computational complexity of SPIG-AwTV is $O(n^2)$, which is consistent with the algorithm complexity of PIBR.

IV. EXPERIMENTS AND RESULTS

To verify the effectiveness of the proposed reconstruction algorithm for suppressing aliasing artifacts and enhancing the quality of OTCL images, both numerical simulations and **real experiments** were performed in this section. The experiment utilized an Intel (R) Core (TM) i7-10700 CPU @ 2.90GHz processor paired with an NVIDIA GeForce RTX 1050Ti GPU. The computational environment was MATLAB R2019a with CUDA 11.1. For leveraging acceleration through multi-threaded forward and backward projection calculations with the TIGRE toolkit [36]. The performance of the reconstruction algorithms is evaluated quantitatively by structural similarity index (SSIM), root mean square error (RMSE), peak signal-to-noise ratio (PSNR), and correlation coefficient (CC).

A. Simulation experiment

The 3D simulated PCB model is shown in Fig. 4(a), with a size of $512 \times 512 \times 30$ pixels and an object size of $10.24\text{mm} \times 10.24\text{mm} \times 0.6\text{mm}$. The PCB model incorporates three different routing layers, each of which is with 4 voxels in thickness. The gaps between the routing layers are 6 voxels. To evaluate the accuracy of various reconstruction algorithms in defect detection, the simulated PCB model includes defects such as through holes and fractures.

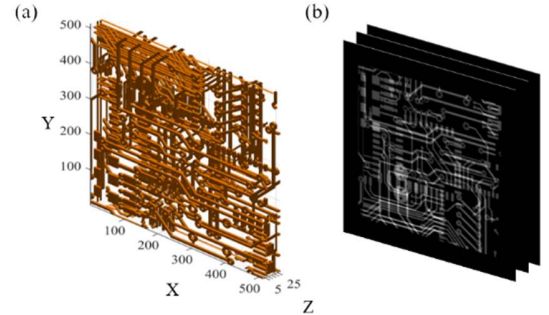


Fig. 4. Simulation model and projection, (a) 3D simulated PCB model, (b) projection diagram.

TABLE I
SIMULATION EXPERIMENT PARAMETERS

Parameters	Value
Distance between source and object h (mm)	50
Distance between detector and object l (mm)	150
Scanning angle range θ_i	$[-\pi/4, \pi/4]$
Number of projection view	11×2
Pixel number of detector (pixel)	1024×1024
Pixel size of detector (mm)	0.05×0.05
Pixel number of object	$512 \times 512 \times 30$

The simulation experiment parameters are shown in Table I. The vertical distance between the source and the object (h) is 50 mm, and the distance from the object to the detector (l) is 150 mm. The detector has 1024×1024 pixels, with a pixel size of 0.05 mm. The equivalent projection angle range of

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OTCL is $\theta_i = [-\pi/4, \pi/4]$, and 11 projection views are collected at equal angles for each segment. The projection diagram is depicted in Fig. 4(b).

B. Noise-free simulation results

In the noise-free case, the proposed algorithm SPIG-AwTV is compared with CLFBP, simultaneous iterative reconstruction technique (SIRT), simultaneous algebraic reconstruction technique-total variation (SART-TV), SART-TAwTV and PIBR algorithms. For iterative algorithms, a series of parameters are tested and the parameters with the best results are chosen to ensure fairness in comparison. The number of iterations for SIRT is set as 300, the TV

regularization coefficient of SART-TV is set as 0.002, the iteration number for TV is 50. The parameters of SART-TAwTV is adjusted to reference [19]. The parameters of the PIBR algorithm are mainly based on reference [26]. The parameters of the proposed SPIG-AwTV algorithm are set as: $M_{iter} = 300$, $R_{iter} = 20$, $S_{iter} = 10$, $\mu = 2$, $\alpha = 0.8$, $\beta = 0.8$, $\lambda = 1$, $\lambda_1 = 0.005$, $\sigma = 0.008$.

Fig. 5 displays the results of various algorithms under noise-free conditions, depicting 3D images, the 6th, 16th, and 28th layers of the X-Y plane and the 150th layer of the X-Z plane. Fig. 6 presents localized magnifications of the three regions of interest (ROI) identified in Fig. 5.

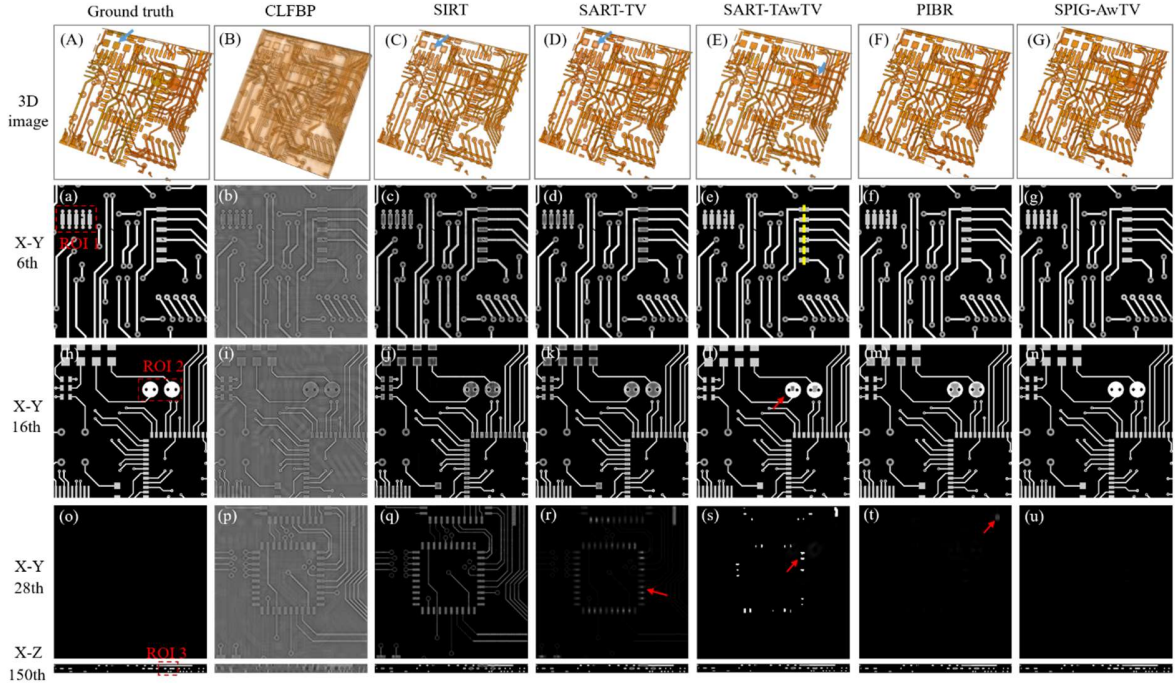


Fig. 5. Comparison of results under noise-free condition. The first row shows the 3D images, the second to fourth row show the results of 6th, 16th and 28th layers, respectively. The last row is 150th in X-Z plane. Each column from left to right are the results of Ground truth, CLFBP, SIRT, SART-TV, SART-TAwTV, PIBR, and SPIG-AwTV. The display windows of (a)-(n) are [0, 1], while (o)-(u) are [0, 0.1].

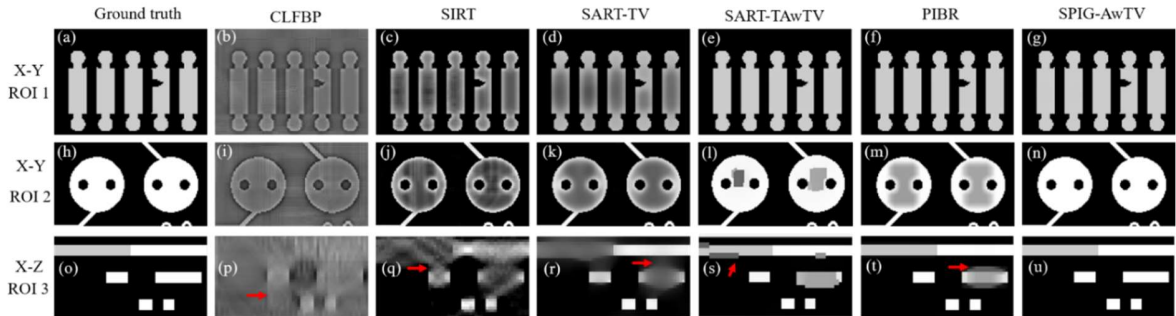


Fig. 6. Zoom in ROIs in Fig. 5. The first row shows the ROI1s of by different methods, the second row shows the ROI2s, the third row shows the ROI3s. Each column from left to right is the result of Ground truth, CLFBP, SIRT, SART-TV, SART-TAwTV, PIBR, and SPIG-AwTV.

The area indicated by the blue arrow in 3D images shows the differences between different algorithms. Observing the results of the X-Y plane in Fig. 5: the layered architecture of

the CLFBP algorithm is evident, but it is accompanied by aliasing artifacts resulting from interference between layers. In comparison to the CLFBP algorithm, the SIRT algorithm has

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notably reduced aliasing artifacts, albeit orbital issues regarding grayscale balance and contrast are apparent. The SART-TV, SART-TAwTV and PIBR algorithms improve the quality of OTCL reconstructed images by introducing different prior information, **but there are still varying degrees of artifacts present.**

A closer examination of the magnified region of interest depicted in Fig. 6, the grayscale uniformity of the reconstructed image by SART-TV algorithm is still suboptimal. The SART-TAwTV algorithm further enhances the non-uniformity of the image, but reveals noticeable blocky artifacts. The PIBR reconstruction shows suboptimal reconstruction result, but with inconsistent grayscale distribution (as seen in ROI 2). **In contrast, the SPIG-AwTV method demonstrates its efficacy in preserving edges, suppressing inter-slice aliasing, and maintaining image contrast.**

Observing the reconstruction results of the 150th layer in the X-Z plane in Fig. 5 and Fig. 6, there is a significant structural drag phenomenon in the CLFBP and SIRT reconstruction results. **It should be noted that the aliasing artifacts exhibited by the X-Y plane and the dragging artifacts exhibited by the X-Z plane are defined by observing the reconstructed image from different directions, and both are essentially consistent.** The reconstruction images of SART-TV and PIBR algorithms suppress most of the structural dragging phenomenon, but there still have errors in the area indicated by the red arrow in ROI3. The reason is traditional TV uses the same weight in all directions, so it cannot effectively avoid the diffusion of corresponding aliasing artifacts. **SPIG-AwTV has dual constraints of its own prior contour and more advanced TV term, thus it can improve the resolution in the Z direction and avoid structural drafting.**

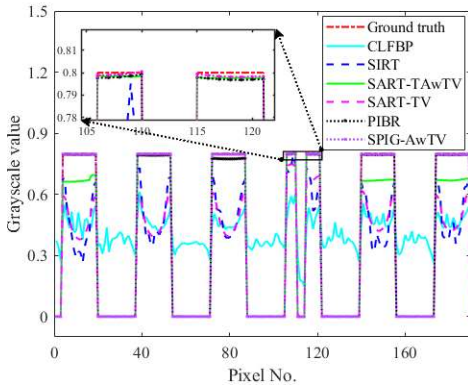


Fig. 7. Horizontal profiles of reconstructed images along the yellow dashed line plotted in Fig. 5.

To provide a more detailed comparison of the related algorithms, the profiles of the reconstructed images (yellow dashed line plotted in Fig. 5(e)) are shown in Fig. 7. Obviously, the grayscale distribution curve of SPIG-AwTV closely matches the Ground truth, which further confirms that SPIG-AwTV can effectively eliminate aliasing artifacts and ensure accurate grayscale values.

The values of quantitative evaluation indicators are given in

Table II. The metrics of SPIG-AwTV perform the best result. The PSNR reaches 42.9045, and increase of about 20% compared to SART-TAwTV and PIBR. The numerical evaluation indicators indicate that the image quality reconstructed by the SPIG-AwTV algorithm is superior.

TABLE II
QUANTITATIVE EVALUATION OF RECONSTRUCTIONS UNDER NOISE-FREE CONDITION

	PSNR	RMSE	SSIM	CC
CLFBP [30]	7.9637	0.3998	0.0868	0.5674
SIRT	20.6858	0.0924	0.8984	0.9168
SART-TV [9]	26.2154	0.0489	0.9739	0.9791
SART-TAwTV [19]	34.2621	0.0194	0.9956	0.9957
PIBR [26]	35.5207	0.0167	0.9972	0.9968
SPIG-AwTV	42.9045	0.0072	0.9993	0.9990

To analyze the convergence of iterative algorithms, Fig. 8 displays the variation curve of RMSE during the iteration process. As the number of iterations increases, all reconstruction algorithms tend to converge. Comparing with other algorithms, the results show that the SPIG-AwTV converges to a smaller RMSE in noise-free cause.

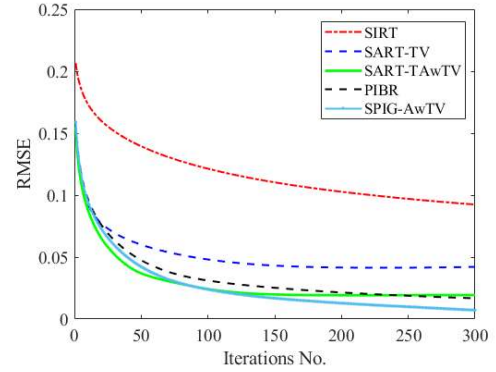


Fig. 8. RMSE curves of the reconstructed workpiece phantom during iterations.

TABLE III
THE ITERATION TIME OF THE ITERATIVE RECONSTRUCTION ALGORITHM FOR EACH ITERATION

Algorithm	Time /per (s)	Algorithm	Time /per (s)
SIRT	1.2	PIBR	18.8
SART-TV	7.4	SPIG-AwTV	16.7
SART-TAwTV	7.7		

The per iteration time of the iterative reconstruction algorithm is shown in Table III. PIBR and SPIG-AwTV naturally have longer time consumption than other algorithms due to the utilization of TV regularization and prior image information. Compared with PIBR, SPIG-AwTV avoids the process of obtaining prior information and image registration, and can provide better image quality.

In conclusion, the qualitative and quantitative analysis of the reconstruction results under noise-free conditions demonstrated that the proposed SPIG-AwTV algorithm for OTCL image reconstruction exhibits substantial advantages in

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aliasing artifact reduction, edge integrity.

C. Noise simulation results

To verify the robustness of the proposed method to noise, noisy simulation experiments were conducted. Poisson noise and Gaussian noise were added to the projection data of OTCL, and the noise model can be represented as [37]:

$$p_{noise} = -\log\left(\frac{Poisson(I_0 \times \exp(-p)) + Gaussian(0, \sigma_e^2)}{I_0}\right) \quad (28)$$

Specifically, p represents the noise-free data, and p_{noise} represents the projection data with noise, $I_0 = 6000$ (I_0 denotes the number of incident photons before attenuation), the mean value of Gaussian noise is zero and the variance $\sigma_e^2 = 1$.

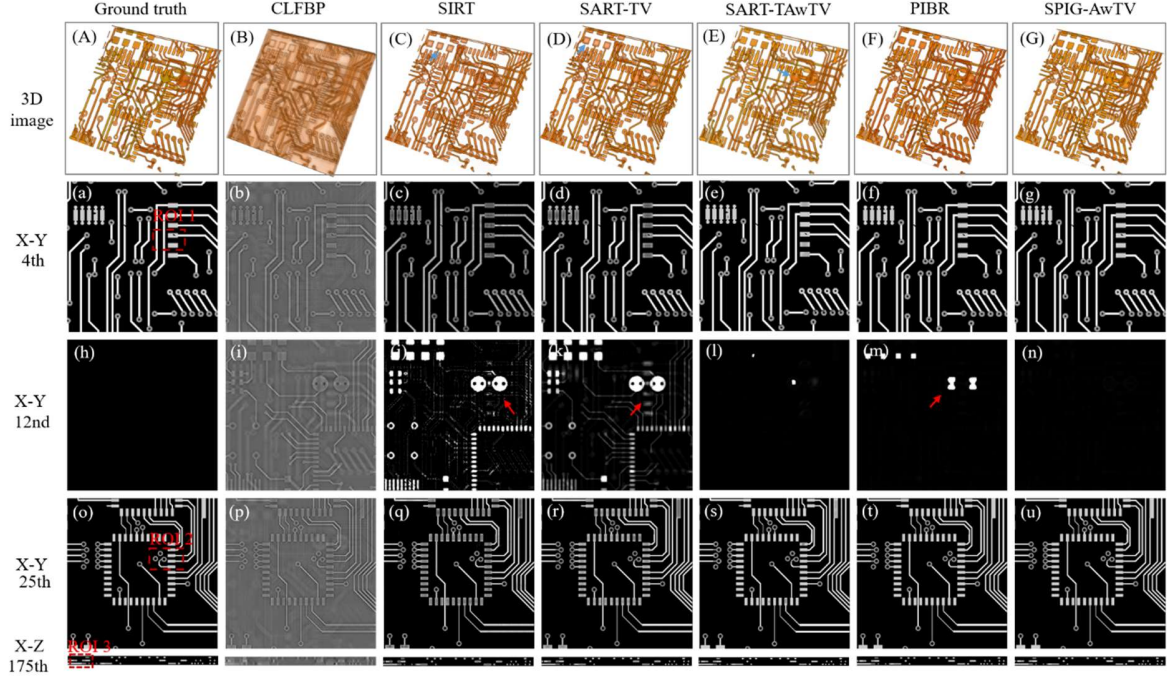


Fig. 9. Simulation experiment results under noisy condition. The first row shows the 3D images, the second to fourth rows show the results of 4th, 12nd and 25th layers, respectively. The last row is 175th in X-Z plane. Each column from left to right is Ground truth, CLFBP, SIRT, SART-TV, SART-TAwTV, PIBR and SPIG-AwTV. The display window of third row is $[0, 0.1]$, while the rest are $[0, 1]$.

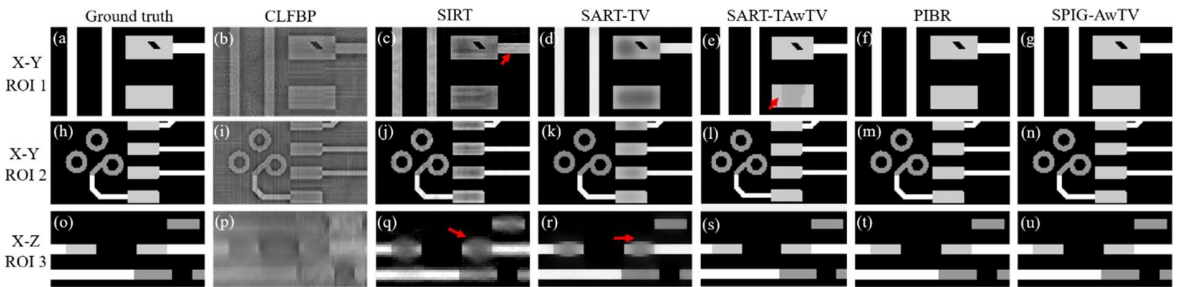


Fig. 10. Zoom-in views of ROIs in Fig. 9. The first row shows the ROI1s of the reconstructed results from noisy data by different methods, while the second row shows the ROI2s of the results. Each column from left to right is the result of Ground truth, CLFBP, SIRT, SART-TV, SART-TAwTV, PIBR and SPIG-AwTV. The display gray scale is $[0, 1]$.

In the noisy case, after adjusting the parameters, the parameters of SPIG-AwTV are set as $R_{iter} = 25$, $S_{iter} = 8$, $\lambda_1 = 0.004$ and the remaining parameters mirroring those under noise-free conditions. Observing Fig. 9, it can be seen that the reconstructed image from the CLFBP algorithm is heavily affected by aliasing artifacts. Especially for the 12nd image in the X-Y plane, it is not only accompanied by severe aliasing

artifacts but also obvious noise. The SIRT algorithm struggled to produce satisfactory reconstruction results due to the absence of introduced prior information. In contrast to the CLFBP reconstruction, the image quality of SART-TV, SART-TAwTV and PIBR reconstruction has been significantly improved, but there are still aliasing artifacts as shown by the red arrow. Leveraging the constraints of self-prior information and adaptive weighted TV, the SPIG-AwTV

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algorithm proposed in this study achieved superior image quality across various layers, with nearly negligible aliasing artifacts and noise.

Fig.10 illustrates the zoom-in views of ROIs in Fig. 9. From the enlarged ROI images in the X-Y plane, it can be found that the results of CLFBP and SIRT accompanied by significant noise. SART-TV and SART-TAwTV have almost no noise, but there is uneven grayscale. The SPIG-AwTV algorithm obtains more accurate vertical edges and has a superior ability to aliasing suppression and noise suppression. Additionally, observing the enlarged ROI images in the X-Z plane, the result of SPIG-AwTV algorithm shows clear interlayer structures, free from structural dragging or noise.

TABLE IV

QUANTITATIVE EVALUATION OF RECONSTRUCTIONS UNDER NOISE CONDITION

	PSNR	RMSE	SSIM	CC
CLFBP [30]	7.9575	0.4001	0.0863	0.5660
SIRT	20.6718	0.0926	0.8980	0.9165
SART-TV [9]	26.1357	0.0493	0.9733	0.9788
SART-TAwTV [19]	34.1549	0.0196	0.9956	0.9956
PIBR [26]	34.3344	0.0171	0.9944	0.9967
SPIG-AwTV	41.1070	0.0078	0.9970	0.9989

For quantitative evaluation, the metrics of the reconstructed images under noise are shown in Table IV. The images reconstructed by SPIG-AwTV algorithm achieved the best values among all evaluation metrics. Compared with other iterative algorithms, the SPIG-AwTV algorithm has stronger robustness and noise resistance to noise.

D. Influence of the self-prior information

In this section, the influence of self-prior information on the proposed SPIG-AwTV algorithm is analyzed under different self-prior weight values λ_1 . The other parameters of SPIG-AwTV remain consistent and are all set as: $M_{iter} = 300$, $R_{iter} = 20$, $S_{iter} = 10$, $\mu = 2$, $\alpha = 0.8$, $\beta = 0.8$, $\sigma = 0.008$. The geometric parameters of the experimental data are shown in Table 1.

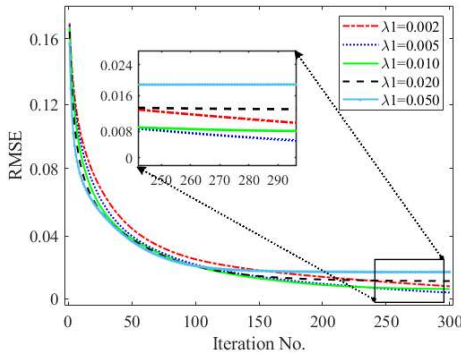


Fig. 11. The RMSE curves of proposed SPIG-AwTV algorithm under different self-prior weight values λ_1 .

Fig. 11 depicts the quantitative results for different λ_1 . Generally, the larger λ_1 leads to a faster convergence speed but the convergence value may not be optimal due to the prior

information may have some structural inaccuracies. Therefore, it is necessary to consider both convergence speed and convergence value comprehensively.

Table V shows the quantitative results corresponding to the reconstruction results under different self-prior weight. Comparing quantitative evaluation values, when the weight λ_1 is 0.005, SPIG-AwTV achieved the best quantitative indicators. When λ_1 is too large, the indicators of SPIG-AwTV images have significantly decreased. Therefore, the setting of λ_1 needs to consider the accuracy of the self-prior information.

TABLE V

QUANTITATIVE EVALUATION THE IMPACT OF λ_1 ON THE RECONSTRUCTION RESULTS

	PSNR	RMSE	SSIM	CC
$\lambda_1 = 0.002$	36.7044	0.0108	0.9990	0.9986
$\lambda_1 = 0.005$	42.9045	0.0072	0.9993	0.9990
$\lambda_1 = 0.010$	38.8452	0.0093	0.9989	0.9989
$\lambda_1 = 0.020$	33.6188	0.0139	0.9978	0.9981
$\lambda_1 = 0.050$	30.0431	0.0191	0.9957	0.9968

From the above reconstruction results, it can be found that the weight of self-prior information has a significant impact on the convergence speed and reconstruction quality. Therefore, in order to ensure the quality of the reconstructed images by SPIG-AwTV algorithm, it is necessary to choose an appropriate self-prior weight value.

E. Influence of the AwTV

The SPIG-AwTV contains an adaptive weighted TV regularization term (i.e. AwTV), which is controlled by σ . σ is used to adjust the amount of smoothing applied to edge voxels relative to non-edge voxels. Theoretically, the smaller σ gives low weights to almost all voxels, making the model inefficient in artifact-suppressing. When $\sigma \rightarrow \infty$, the weight value will approach to 1, and the AwTV will be similar to traditional TV.

To assess the impact of AwTV parameters on the reconstruction outcomes, the reconstructed results under different values of σ are compared.

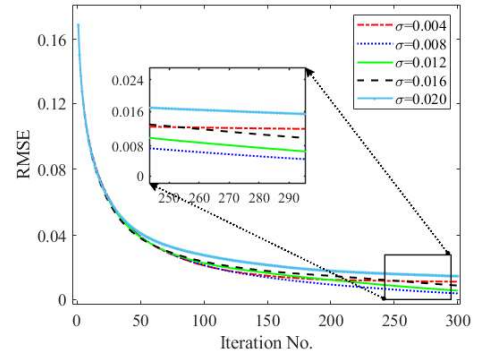


Fig. 12. The RMSE curves of proposed SPIG-AwTV algorithm under different weight values of σ .

Fig. 12 displays the RMSE curves of the proposed SPIG-AwTV algorithm under different values of σ . When $\sigma = 0.008$, the SPIG-AwTV achieved the minimum RMSE value, and the

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corresponding quantitative analysis indicators were also optimal (as shown in Table VI).

TABLE VI
QUANTITATIVE EVALUATION THE IMPACT OF σ ON THE
RECONSTRUCTION RESULTS

	PSNR	RMSE	SSIM	CC
$\sigma = 0.004$	33.5887	0.0139	0.9989	0.9982
$\sigma = \mathbf{0.008}$	42.9045	0.0072	0.9993	0.9990
$\sigma = 0.012$	39.6603	0.0088	0.9990	0.9990
$\sigma = 0.016$	35.6812	0.0117	0.9985	0.9986
$\sigma = 0.020$	31.2378	0.0171	0.9969	0.9975

F. Real experiment results

To analyze the performance of the SPIG-AwTV in actual data, an experimental platform was built as shown in Fig. 13(a). The X-ray source is Japan Hamamatsu L10321, with an X-ray cone angle of 118° . The flat panel detector adopts the IRUI NDT0505J, with pixel size of $85 \mu\text{m} \times 85 \mu\text{m}$ and pixel matrix of 1536×1536 . In this OTCL experimental system, the source is installed below the turntable, emitting a cone beam of X-rays upwards. The detector is installed above the turntable to receive X-rays passing through a plate-shaped object. The orthogonal projection data is obtained by rotating the turntable with an angle $\pi/2$.

TABLE VII
SYSTEM AND GEOMETRICAL SCANNING PARAMETERS

Parameter	Value
$l(\text{mm})$	110
$h(\text{mm})$	30
Scanning angle range θ_i	$[-\pi/4, \pi/4]$
Number of projections	31×2
Pixel number of detector (pixel)	1536×1536
Pixel size of detector (mm)	0.085×0.085
Tube voltage/current (kV/ μA)	80/120

The actual physical experimental parameters are shown in Table VII. The test object is a computer memory module (as shown in Fig. 13(b)), with a physical size of approximately $60 \text{ mm} \times 60 \text{ mm} \times 5 \text{ mm}$, which belongs to a typical plate-like structure. Collecting projection data of computer memory module by the built OTCL experimental platform. The collected projection data is shown in Fig. 13(c),

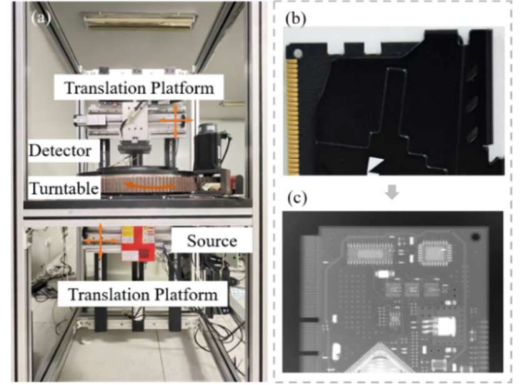


Fig. 13. Physical experimental platform and imaging object. (a) OTCL experimental platform, (b) computer memory module, (c) projection data.

Fig. 14 shows the reconstructed 3D images of the computer memory module, reconstructed by CLFBP, SIRT, SART-TV, SART-TAwTV and SPIG-AwTV algorithms, respectively. The object is reconstructed with $1024 \times 1024 \times 50$ voxels. After adjusting the algorithm parameters, the parameters of SPIG-AwTV are set as: $M_{iter} = 50$, $R_{iter} = 30$, $S_{iter} = 10$, $\mu = 2$, $\alpha = 0.8$, $\beta = 0.8$, $\sigma = 0.008$, $\lambda = 1$, $\lambda_1 = 0.002$. From the Fig. 14, it can be observed that the SPIG-AwTV algorithm can effectively reconstruct images of circuit layers and component layers. On the contrary, other algorithms suffer from aliasing artifacts and structural deficiencies.

Comparing the results of the 33rd (circuit layer) and 45th (component layer) layers in Fig. 15, it can be observed that the images reconstructed by CLFBP are accompanied by obvious aliasing artifacts. Such as, in the reconstruction results of 45th, the components and circuits are basically mixed together, with a low degree of differentiation, and the reconstructed images are accompanied by certain noise. Compared to CLFBP reconstruction results, SIRT increases the contrast of the image, but it still has aliasing artifacts as indicated by the red arrows. The SART-TV algorithm has better reconstruction result than the SIRT algorithm, but there is still a lack of contrast in the area indicated by the red arrow. The 33rd layer image reconstructed by SART-TAwTV has structural defects (as shown by the red arrow in Fig. 15 (d)). The proposed SPIG-AwTV has the highest image contrast and the best reconstruction results in the above OTCL algorithms, which substantially reducing aliasing artifacts.

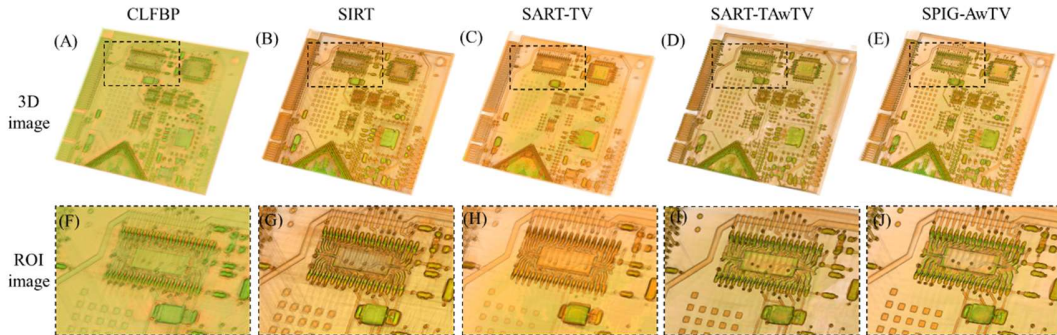


Fig. 14 Computer memory module 3D reconstruction results. The first row shows the results of CLFBP, SIRT, SART-TV, SART-TAwTV and SPIG-AwTV, respectively. The second row is the corresponding ROI images.

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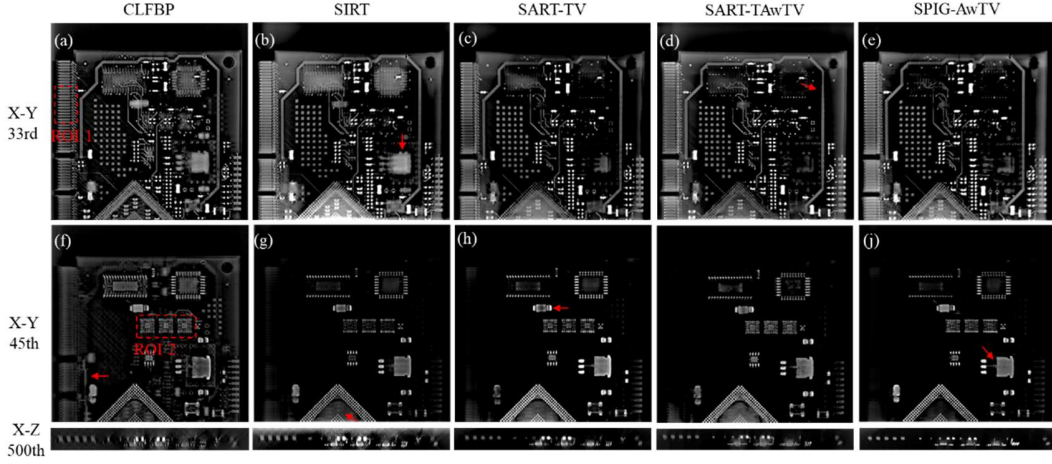


Fig. 15 Computer memory module results. The first row shows the 33rd reconstructed results in X-Y plane by different methods, the second row shows the 45th of the results in X-Y plane, while the third row shows the 500th of the results in X-Z plane. Each column from left to right is the result of CLFBP, SIRT, SART-TV, SART-TAwTV and SPIG-AwTV. The display window is [0.05, 0.9].

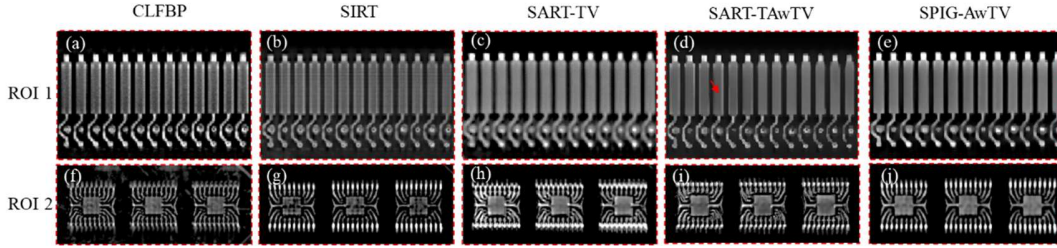


Fig. 16 Enlarged images of ROIs. The first row shows the results of ROI1, and the second row shows the results of ROI2. Each column from left to right is the result of CLFBP, SIRT, SART-TV, SART-TAwTV and SPIG-AwTV. The display window is [0.05, 0.9].

Although the proposed SPIG-AwTV algorithm cannot completely suppress aliasing artifacts in the reconstructed image, observing the locally enlarged images of ROI1 and ROI2 in Fig. 16, the image reconstructed by SPIG-AwTV algorithm with a clearer structure and little noise. Analyzing the experimental results, it can be found that the ability of the SPIG-AwTV algorithm to suppress aliasing artifacts is related to the self-prior information obtained. The self-prior information extracted by the SPIG-AwTV reconstruction algorithm depends on the mask structure obtained by CLFBP. If the obtained mask structure cannot fully cover the structure, it will lead to the loss of some structural information on the X-Y plane. Conversely, an excessively large mask structure can diminish the suppression effect on aliasing artifacts. Therefore, further research can optimize the method of extracting mask structure, and the mask structure can be continuously optimized during the iteration process to improve the suppression effect of aliasing artifacts.

V. DISCUSSION AND CONCLUSION

To suppress aliasing artifacts and enhance the quality of OTCL images, the frequency domain characteristics of OTCL was analyzed, and a SPIG-AwTV algorithm was proposed. SPIG-AwTV initially extracts self-prior information from CLFBP results, which is utilized to form a self-prior regularization term (SPIG). Concurrently, considering the

gradient differences, an AwTV regularization term is designed to further constrain the reconstruction results. The proposed SPIG-AwTV algorithm demonstrates superior performance in suppressing aliasing artifacts and noise compared to CLFBP, SIRT, SART-TV, SART-TAwTV and PIBR algorithms.

Notably, SPIG-AwTV provides an OTCL image reconstruction algorithm framework, which does not necessitate additional imaging steps for acquiring prior information, thus avoiding the need for registration of prior information relative to reconstructed images. Meanwhile, a method for constructing self-prior regularization term by self-prior information was also proposed. Subsequent research can modify the regularization terms to form new reconstruction algorithms.

By analyzing the 3D frequency domain distribution of OTCL, it was observed that there is a lack of data extending vertically from the central slice, corresponding to the vertex position. The incomplete data leads to artifacts in the reconstructed OTCL images. For the layer surface (X-Y plane), the above issues result in aliasing artifacts and loss of image contrast. For the transverse plane (X-Z plane), the missing frequency domain data cause dragging artifacts. The concepts of two types of artifacts are generated by analyzing OTCL reconstructed images from different directions, and their essence is consistent. Therefore, suppressing aliasing artifacts and improving the image quality of the layer surface can also enhance the image quality of the transverse plane.

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Similar to the frequency domain distribution of OTCL, RCL also exhibits missing conical data in the 3D frequency domain space, centered on the vertex and extending in both upward and downward directions. The mathematical essence problems faced by OTCL and RCL image reconstruction are consistent. Therefore, the SPIG-AwTV algorithm is also applicable to RCL image reconstruction.

Several challenges require further investigation. Notably, the self-prior constrained algorithms in OTCL reconstruction present a particular concern, while the employment of a self-prior is promising in mitigating aliasing artifacts and enhancing the reconstructed image quality, its accuracy and weight settings may compromise the result. Therefore, it is necessary to improve self-prior acquisition and adaptive weighting. Moreover, exploring methods to effectively integrate self-prior information into the OTCL reconstruction is also essential for improving the quality of the reconstructed images.

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